MA1522 Linear Algebra for Computing

Week 5 Problem Solving Session

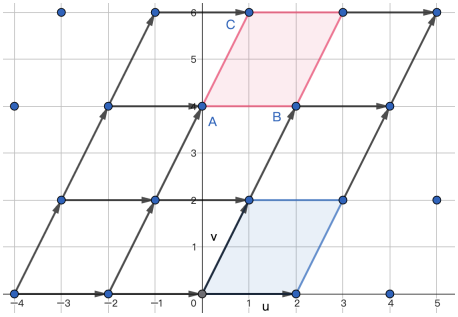
Announcements

1. Attempt quiz 2.8 to 3.2, due on Friday 12 September 2025.
2. Attempt quiz 3.3 to 3.5, due on Friday 19 September 2025.
3. Homework 1 published. Due Sunday, 14 September 2025. Download your submitted file to check that it is the right, complete, file, and all the pages are clear. Canvas will impose the late penalty automatically for any submission after deadline.
4. Midterm on 9 October 2025, 1900hrs to 2000hrs at MPSH (venue and seating plan to be announced later). Indicate your availability in Canvas https://canvas.nus.edu.sg/courses/79402/quizzes/66316.

Submission

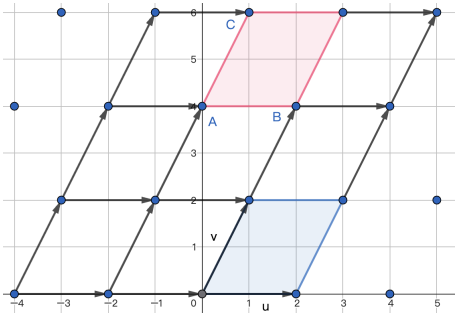
- The questions are available in Canvas Assignment, with copyable MATLAB codes.
- Submit your solutions only for those labeled **Problem 1 to 3**.
- If handwritten on A4 paper: scan or take clear photos of your work. Merge all pages into a single PDF file, arranged in the correct order.
- If typed or written digitally: save or export directly as a PDF file.
- Name your file as **StudentNo_week5** (e.g. **A1234567Z_week5.pdf**).
- Upload your PDF file to Canvas under *Lecture group 2: Problem Solving Session Week 5*.
- **Deadline: Today, 1830hrs.**

Problem 1(a)



Let $\mathbf{u} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Express $\mathbf{A} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$, $\mathbf{C} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$ in terms of $s\mathbf{u} + t\mathbf{v}$ for some $s, t \in \mathbb{R}$.

Problem 1(b)

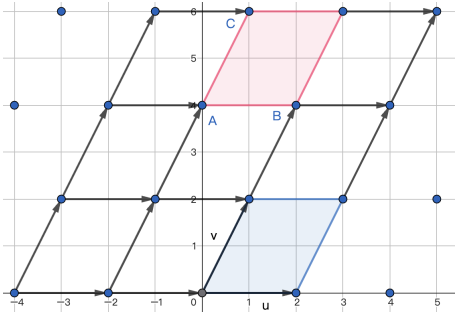


Express the blue and red parallelograms as

$$\{s\mathbf{u} + t\mathbf{v} \mid a \leq s \leq b,\ c \leq t \leq d\}$$

for some $a, b, c, d \in \mathbb{R}$ respectively.

Problem 1(c)



Explain why $\mathrm{span}\{\mathbf{u}, \mathbf{v}\} = \mathbb{R}^2$ from the picture.

Problem 2

Let $\mathbf{u}_1 = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 3 \end{pmatrix}$, $\mathbf{u}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \\ -1 \end{pmatrix}$, $\mathbf{u}_3 = \begin{pmatrix} 2 \\ -1 \\ -1 \\ 1 \end{pmatrix}$. Find the set of vectors $\mathbf{v} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$ such that $\mathrm{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{v}\} = \mathbb{R}^4$.

Store `u1=[-1;1;-1;3]; u2=[2;1;1;-1]; u3=[2;-1;-1;1];`

Problem 3(a)

Determine whether $S = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x = y \right\}$ is a subspace of $\mathbb{R}^3$.

Problem 3(b)

Determine whether $S = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x^2 = y^2 \right\}$ is a subspace of $\mathbb{R}^3$.

Problem 3(c)

Determine whether $S = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x^3 = y^3 \right\}$ is a subspace of $\mathbb{R}^3$.